

Example 4.10 Many of the following are addressed in HW #2

(1) H^0 counting measure

$$H^0(A) = \begin{cases} \# A & \text{if } A \text{ finite,} \\ +\infty & \text{---} \end{cases}$$

$\# A \dots$ cardinality of A

(2) $H^N_0 = (\mathbb{L}^N)^*$

In particular, they have the same measurable sets, so $H^N = \mathbb{L}^N$
Consider $N > 1$.

Use Isodiametric Theorem (Evans & Gariepy Thm 2.4)

$$\forall A \subseteq \mathbb{R}^N \quad (\mathbb{L}^N)^*(A) \leq \alpha_N \left(\frac{\text{diam } A}{2} \right)^N$$

If $\text{diam } A = 2$, then $(\mathbb{L}^N)^*(A) \leq \alpha_N = (\mathbb{L}^N)^*(B(0,1))$.

so the unit ball has the largest "volume" among all sets with same diameter 2.

let $\delta > 0$ let $A \subseteq \bigcup_{n=1}^{\infty} E_n$, $\text{diam } E_n < \delta$.

$$\text{Subadditivity} \Rightarrow (\mathbb{L}^N)^*(A) \leq \sum_{n=1}^{\infty} (\mathbb{L}^N)^*(E_n)$$

$$(\text{Isodiametric} \leq) \leq \sum_{n=1}^{\infty} \alpha_N \left(\frac{\text{diam } E_n}{2} \right)^N$$

Take inf over all such covers, get

$$(\mathbb{L}^N)^*(A) \leq H^N_{\delta}(A)$$

let $\delta \rightarrow 0^+$ and conclude that

$$(\mathbb{L}^N)^*(A) \leq H^N_0(A).$$

Conversely: $C_N := \alpha_N \left(\frac{\sqrt{N}}{2} \right)^N.$

$R \subseteq \mathbb{R}^N$ cube $\alpha_N \left(\frac{\text{diam } R}{2} \right)^N = C_N \mathcal{L}^N(R).$

If $\delta > 0$ then $H_\delta^N(A) \leq \inf \left\{ \sum_{i=1}^{\infty} \alpha_N \left(\frac{\text{diam } Q_i}{2} \right)^N : Q_i \text{ cubes,} \right.$

$\text{diam } Q_i \leq \delta, A \subseteq \bigcup_{i=1}^{\infty} Q_i \}$

$= C_N \inf \left\{ \sum_{i=1}^{\infty} \mathcal{L}^N(Q_i) : Q_i \text{ cubes, diam } Q_i \leq \delta, A \subseteq \bigcup_{i=1}^{\infty} Q_i \right\}.$

Prop 2.11 δ

Any rectangle can be approximated in $(\mathcal{L}^N)^*$ -measure by rectangles with rational sides lengths, and in turn, each rectangle with side lengths rational can be decomposed as unions of cubes.

$= C_N (\mathcal{L}^N)^*(A).$

let $\delta \rightarrow 0^+$ and get $H_\delta^N(A) \leq C_N (\mathcal{L}^N)^*(A) \quad (4.3^{***})$

↑ ① ?

Thm [Covering Thm]

let $\mathcal{F} \subseteq \mathcal{D}(\mathbb{R}^n)$ be a collection of (nondegenerate) closed balls. let $\tilde{A}$ denote the set of centers of the ball that belong to $\mathcal{F}$, and assume that $(\mathcal{L}^n)^*(\tilde{A}) < \infty$ and $\inf \{ r > 0 : B(a, r) \in \mathcal{F} \} = 0 \quad \forall a \in \tilde{A}.$



If $U \subseteq \mathbb{R}^n$ is an open set, then there exists a countable collection of disjoint ball in $\mathcal{F}$ s.t.

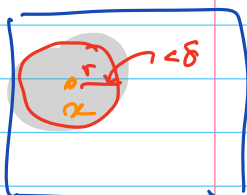
$\bigcup_{B \in \mathcal{G}} B \subset U \quad \text{and} \quad (\mathcal{L}^n)^* \left((\tilde{A} \cap U) \setminus \bigcup_{B \in \mathcal{G}} B \right) = 0.$

Fix $\varepsilon, \delta > 0$ and let R_i be rectangles st. for $A \subseteq \mathbb{R}^N$

$$A \subseteq \bigcup_{i=1}^{\infty} R_i, \quad \text{diam } R_i < \delta, \quad \sum_{i=1}^{\infty} \mathcal{L}^N(R_i) \leq (\mathcal{L}^N)^*(A) + \varepsilon.$$

$\bullet$
 R_i

$\bullet$
 $\dot{R}_i \dots$ interior of R_i



write $\dot{R}_i = \bigcup_{\substack{x \in \dot{R}_i \\ r < \delta/2, B(x, r) \subset \dot{R}_i}} \overline{B(x, r)} = \bigcup \text{balls}$

$\dot{A}_i = \{ \text{centers of all these balls} \} \subset \dot{R}_i, \quad (\mathcal{L}^N)^*(\dot{A}_i) < \infty$

Covering Thm $\Rightarrow \exists$ disjoint closed balls $\{B_k^i\}_{k=1}^{\infty} \subset \dot{R}_i$ s.t.

$$(\mathcal{L}^N)^*(R_i \setminus (\bigcup_{k=1}^{\infty} B_k^i)) = (\mathcal{L}^N)^*(\dot{R}_i \setminus (\bigcup_{k=1}^{\infty} B_k^i)) = 0. \quad (4.3^{***})$$

By $(4.3^{**}) \Rightarrow H_0^N(R_i \setminus (\bigcup_{k=1}^{\infty} B_k^i)) = 0.$

We have

$$\begin{aligned} H_{\delta}^N(A) &\leq \sum_{i=1}^{\infty} H_{\delta}^N(R_i) \\ &= \sum_{i=1}^{\infty} \left[H_{\delta}^N\left(\bigcup_{k=1}^{\infty} B_k^i\right) + \underbrace{H_{\delta}^N\left(R_i \setminus \left(\bigcup_{k=1}^{\infty} B_k^i\right)\right)}_{=0} \right] \\ &\leq \sum_{i=1}^{\infty} \sum_{k=1}^{\infty} H_{\delta}^N(B_k^i) \end{aligned}$$

$$\leq \sum_{i=1}^{\infty} \sum_{k=1}^{\infty} \alpha_N \left(\frac{\text{diam } B_k^i}{2} \right)^N$$

$$= \sum_{i=1}^{\infty} \sum_{k=1}^{\infty} \mathcal{L}^N(B_k^i) = \sum_{i=1}^{\infty} \underbrace{\mathcal{L}^N\left(\bigcup_{k=1}^{\infty} B_k^i\right)}_{\text{pairwise disjoint}}$$

$$(4.3^{***}) \quad \sum_{i=1}^{\infty} \mathcal{L}^N(R_i) \leq (\mathcal{L}^N)^*(A) + \varepsilon. \quad \text{let } \varepsilon \rightarrow 0, \quad \delta \rightarrow 0.$$

(3) $H_0^s \equiv 0$ if $s > N$ in $\mathbb{R}^N$

Subdivide the unit cube $Q = [-1/2, 1/2]^N$ into m^N subcubes of side length $\frac{1}{m}$ and diameter $\frac{\sqrt{N}}{m}$.

Let $\delta = \frac{\sqrt{N}}{m}$. Then

$$H_{\frac{\sqrt{N}}{m}}^s(Q) \leq \sum_{i=1}^{m^N} \alpha_s \left(\frac{\sqrt{N}}{2m} \right)^s = \alpha_s \left(\frac{\sqrt{N}}{2} \right)^s \frac{1}{m^{sN}} \xrightarrow{m \rightarrow \infty} 0$$

$$\Rightarrow H_0^s(Q) = \lim_{\delta \rightarrow 0} H_\delta^s(Q) = 0.$$

$\mathbb{R}^N$ can be written as a countable union of translations of the unit cube $\Rightarrow H_0^N(\mathbb{R}^N) = 0$.

(4) Translation Invariance and Positivity Homogeneity of Degree s

For $E \subseteq \mathbb{R}^N$, $x \in \mathbb{R}^N$, $\lambda > 0$, we have

$$H_0^s(x + E) = H_0^s(E), \quad H_0^s(\lambda E) = \lambda^s H_0^s(E).$$

(5) $E \subseteq \mathbb{R}^N$, $0 \leq s < t < +\infty$.

(a) If $H_\delta^s(E) > 0$ for some $0 < \delta \leq \infty \Rightarrow H_0^s(E) > 0$

(b) $H_0^s(E) < +\infty \Rightarrow H_0^t(E) = 0$

(c) $H_0^t(E) > 0 \Rightarrow H_0^s(E) = \infty$

(6) $0 < s < N$ then the Hausdorff measure H_0^s is not σ -finite, i.e., we cannot write

$$\mathbb{R}^N = \bigcup_{n=1}^{\infty} A_n \quad \text{with} \quad H_0^s(A_n) < \infty \quad \forall n \in \mathbb{N}.$$

Why: If $\exists s \in (0, N)$ s.t. $\mathbb{R}^N = \bigcup_{n=1}^{\infty} A_n$, $H_0^s(A_n) < \infty$ then

$\Rightarrow$ (5(b) with $t=N$) $H_0^N(A_n) = 0$.

By (2) above, $(\mathcal{J}_0^N)^*(A_n) = 0 \Rightarrow$ (Schaeffer's) $(\mathcal{J}_0^N)^*(\mathbb{R}^N) = 0$.

Definition 4.1 The Hausdorff dimension of a set $E \subseteq \mathbb{R}^N$

is defined as $\dim_H(E) := \inf \{ 0 \leq s < \infty : H_0^s(E) = 0 \}$.

Note:

- $\dim_H(E) \in [0, +\infty]$ not necessarily an integer number

- $\dim_H(E) \leq N \quad \forall E \subseteq \mathbb{R}^N$.

Indeed, if $\dim_H(E) > N \Rightarrow \exists s > N$ s.t. $H_0^s(E) > 0$
contradicting Example 4.10(3).

- If $\dim_H(E) = t$ then

$\dots s > t \Rightarrow H_0^s(E) = 0$

Indeed, if $s > t \Rightarrow \exists t' \in (t, s) : H_0^{t'}(E) > 0$

$\Rightarrow$ (5(b)) $H_0^s(E) = 0$.

$\dots 0 \leq s < t \Rightarrow H_0^s(E) = \infty$

(use 5(c)).

Example 4.12 [A Lebesgue Measurable Set that is not Borel]

Consider the Cantor set C [Example 3.15 (b)].

$C \subset [0,1]$ is a Borel set and $\mathcal{L}^1(C) = 0$.

claim $\exists F \subset C$ s.t. F is not Borel,

$\mathcal{L}^n$ is complete (4.10 (i)) and C measurable of measure zero $\Rightarrow F$ Lebesgue-measurable.

In particular, this will show that $\mathcal{L}^n|_{\beta(\mathbb{R}^n)} : \beta(\mathbb{R}^n) \rightarrow [0, \infty]$ is not complete.

Consider the Cantor function (Example 3.15 b))

$f: [0,1] \rightarrow [0,1]$ continuous, non-decreasing

define $g: [0,1] \rightarrow [0,2]$
 $x \mapsto x + f(x)$.

Then g is continuous, strictly increasing, $g([0,1]) = [0,2]$.

$g^{-1}: [0,2] \rightarrow [0,1]$ still continuous.

f is constant on each of the removed subintervals $\Rightarrow$
 g maps those intervals onto intervals of same length $\Rightarrow$

$$[0,1] \setminus C = \bigcup_{n=1}^{\infty} \bigcup_{k=1}^{2^{n-1}} I_{k,n}$$

$\downarrow$ mutually disjoint $\approx I_{k,n}^{n-1}$ "constant"

$$\Rightarrow \mathcal{L}^1(g([0,1] \setminus C)) = \mathcal{L}^1\left(\underbrace{\bigcup_{n=1}^{\infty} \bigcup_{k=1}^{2^{n-1}} g(I_{k,n})}_{\text{mutually disjoint because } g \text{ 1:1}}\right)$$

$$= \sum_{n=1}^{\infty} \sum_{k=1}^{2^{n-1}} \mu^1(I_{k,n}) = \mu^1([0,1] \setminus \mathbb{C}) = 1.$$

$$\text{In turn, } \mu^1(g([0,1] \setminus \mathbb{C}) \cup g(\mathbb{C})) = \mu^1(g([0,1])) \\ = \mu^1([0,2]) = 2$$

$$\Rightarrow 2 = \mu^1(g([0,1] \setminus \mathbb{C}) \cup g(\mathbb{C})) \\ \leq (\mu^1(g([0,1] \setminus \mathbb{C}))) + \mu^1(g(\mathbb{C})) \\ = 1 + \mu^1(g(\mathbb{C})) \\ \Downarrow \\ \mu^1(g(\mathbb{C})) \geq 1.$$

Fact #1 If $A \subseteq \mathbb{R}$ is Lebesgue measurable and $\mu^1(A) > 0$ then $\exists E \subset A$ such that E is not measurable.

(Proof: uses inner and outer regularity of μ^1)

Fact #1 $\Rightarrow \exists E \subseteq g(\mathbb{C})$ non measurable.

Fact #2 • In general, the image of a Borel set under a continuous function is not Borel:

B Borel, h continuous $\not\Rightarrow h(B)$ Borel,

• In general, the image of Lebesgue measurable set under a continuous function is not Lebesgue measurable:

G Lebesgue meas., h continuous $\not\Rightarrow h(G)$ measurable.

$\exists E \subset \mathbb{R}$ Lebesgue meas. s.t. $E+E = \{x+y : x,y \in E\}$ not meas.

• However! If $f: I \rightarrow \mathbb{R}$ is continuous,

$I \subseteq \mathbb{R}$ interval, $E \subset I$ Borel $\Rightarrow f(E)$ is Lebesgue measurable!

More generally:

If X is a complete, separable metric space, $f: X \rightarrow \mathbb{R}$ continuous, $B \subset X$ Borel set then
 $f(B) \subset \mathbb{R}$ is Lebesgue measurable
 (not necessarily Borel). •